Hugo Larios

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Summary:

Mechanical Engineering senior at Cornell University with hands-on experience across propulsion, mechatronic design, and aerodynamic analysis. Progressed from propulsion team member to subteam lead on Cornell's Design Build Fly team, building electrical systems and a static thrust rig for competition aircraft; currently applies FEA and fluid dynamics modeling as a member of the Analytics subteam. Proficient in SolidWorks, Fusion 360, ANSYS (Workbench/Fluent), and Python/C/C++/Java/MATLAB.

Education:

Cornell University, College of Engineering Ithaca, NY
B.S. Mechanical Engineering, GPA 3.33/4.0 *Expected May 2027*
Relevant Coursework:
Propulsion of Aircraft and Rockets, Intermediate Fluids, Intro to Aeronautics, Heat Transfer, and Mechanics of Materials.

Project Team Experience:

Cornell Design Build Fly (CUDBF), Ithaca, NY *September 2024-Present*
Analysis Subteam Member *May 2026-Present*

- Modeled a 14x7e APC propeller on SolidWorks, then simulated airflow past it using ANSYS Fluent. This is being replicated for various propellers, which provides the team with additional data when selecting designs.
- Acting as senior advisor for the Propulsion Subteam, aiding members in preparation for Preliminary Design Review.

Propulsion Subteam Lead *May 2025-May 2026*

- Led a 7-person subteam, organizing weekly tasks and dependencies via Slack and Notion to meet design milestones and prevent overlap between subteams.
- Coordinated internal subteam dependencies to improve development efficiency and manufacturing workflows, shortening iteration time between design and test.
- Utilized a Python and MATLAB sizing script to optimize motor-propeller-battery selections and validated over sixty predictions with experimental thrust data through iterative testing and statistical analysis.
- Modeled integration components, as well as produced drawings and parts lists while documenting manufacturing steps for the end-of-year design report.
- Created three separate test procedures and checklists to improve repeatability, support safe operations, and enable future teams to expand testing processes.

Propulsion Subteam Member *September 2024-May 2025*

- Integrated a flight controller and electrical system into two handmade gliders; used Mission Planner and Ardupilot to automate flight paths and correct control-surface movement via servo tuning.
- Applied machining, crimping, and soldering techniques to assemble electronic components, ensuring high-quality material flow and assembly.
- Assisted with prototype fabrication and assembly of electronic subsystems, ensuring tight tolerances and reliable follow-through on design specifications.

Notable Projects:

Spider Robot, (Inspired by the game TUNIC) *June 2026-Present*

- Designed and modeled a 12-degree-of-freedom quadruped robot (4 legs x 3 servos each) in Fusion 360, developing separate mechanical and computational subsystem assemblies.
- 3D-printed and tested a prototype leg assembly, interfacing servos with a Raspberry Pi and PCA9685 PWM driver to validate joint actuation and walking motion.
- Created a full circuit diagram for the robot and selected all electrical components needed to support a 12-servo actuation and control system.

Static Thrust Rig, (Gathers Propulsion Data) *September 2025-March 2026*

- Designed and built a static thrust test rig with a 7-person team, integrating an HX711 load cell and Arduino to measure thrust output across battery-motor-propeller combinations.
- Expanded instrumentation to include voltage, current, RPM, and throttle sensors for full propulsion characterization.
- Interfaced Arduino with Excel for automated data logging, enabling direct graphical comparison of performance across configurations to inform our competition aircraft propulsion selection.

Skills & Languages:

CAD & Analysis: SolidWorks, Fusion 360, eCalc, ANSYS Workbench, ANSYS Fluent, LabVIEW
Programming: Python, C, C++, Java, MATLAB
Manufacturing: CNC Machining, Milling, 3D Printing, Laser Cutting, Crimping, Soldering
Languages: Fluent in English, Intermediate French, Intermediate Spanish